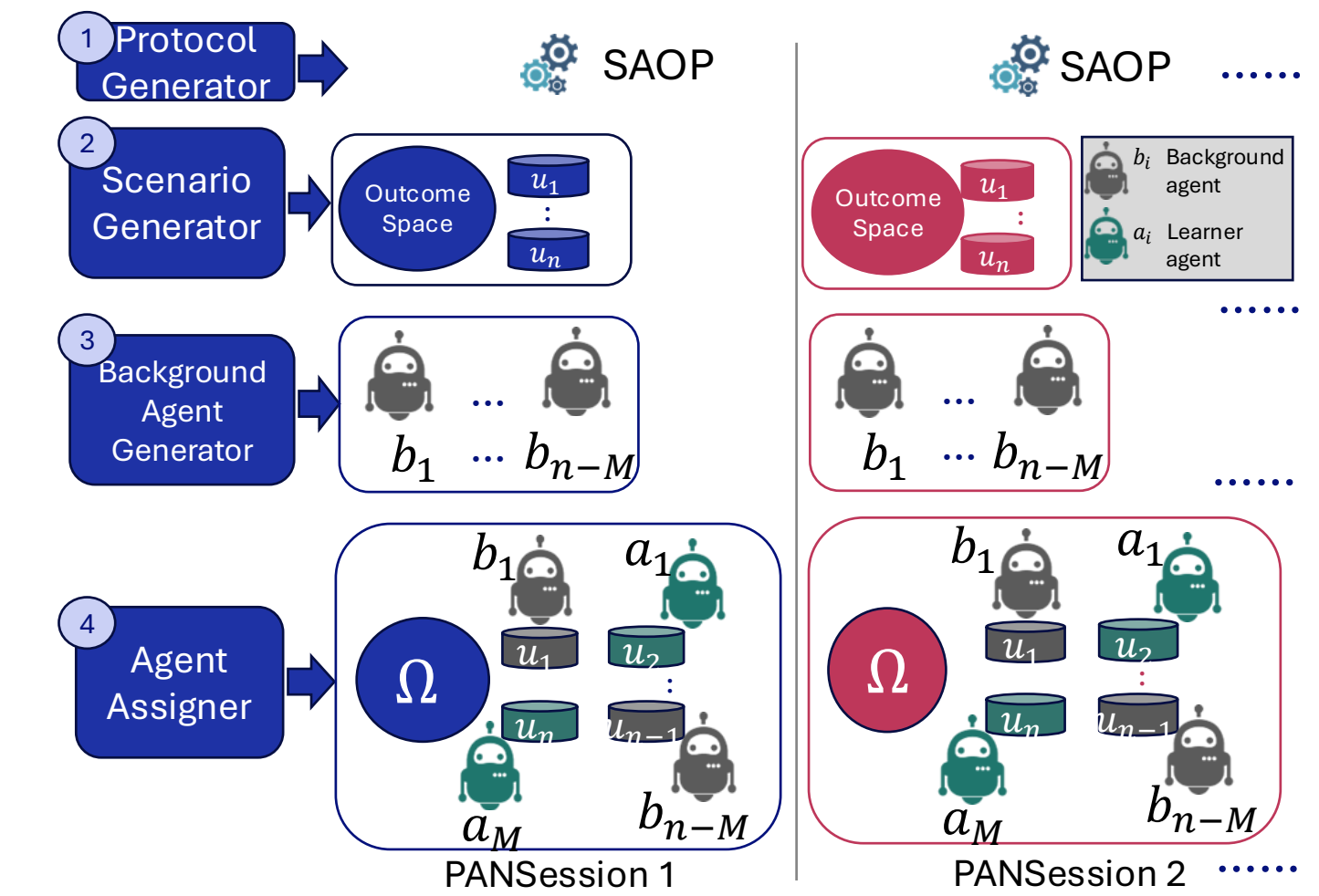


The Question

In RL-based automated negotiation, **how observations and actions are encoded** (the *codec*) decides whether the agent generalizes. Existing codecs force a hard choice: **generalize but discard structure**, or **exploit structure but only on one scenario**.

This Paper

- VUDO (Vector Utility-Dependent Outcome projection): a codec that projects outcomes onto the value-function space of a GLA utility – **generalizes across scenarios and keeps outcome-space structure**, with no feature engineering and no architecture constraints.
- A **unified RL-for-AN framework** (NegMAS-RL): most SOTA methods are special cases of one $(\mathcal{D}, \mathcal{C}, \mathcal{R}, \mathcal{F})$ tuple.
- VUDO **outperforms SOTA RL and heuristic baselines** on hard scenarios – varying utilities, outcome spaces, opponents.



PANSession Generation Process – sample $\mathcal{P}$, λ , backgrounds, assignment, then train / test.

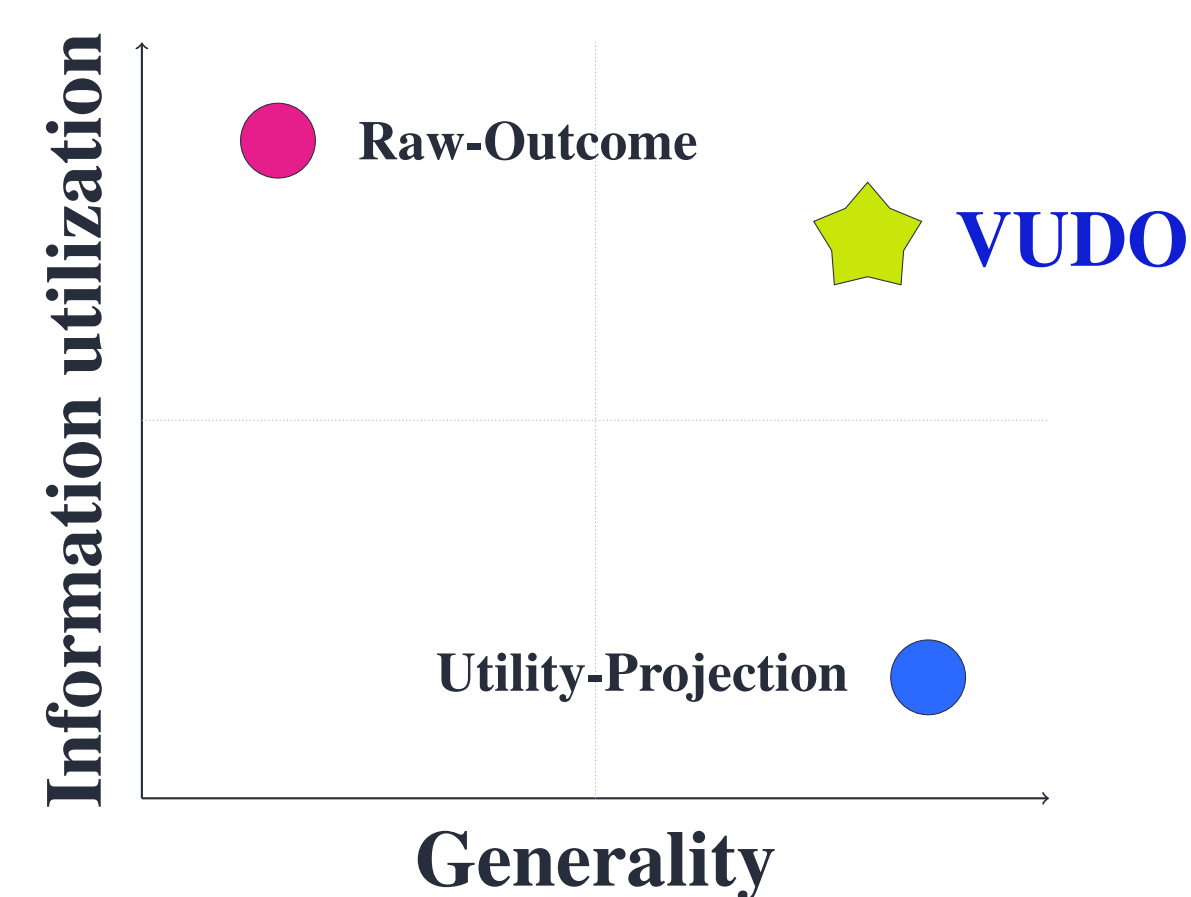
CODECS: THE FORGOTTEN DESIGN CHOICE

With **Generalized Linear Aggregation** utilities, $u(\omega) = \sum_{k=1}^K \alpha_k \nu_k(\omega_{g_k})$, most RL-for-AN systems pick one of two extremes:

Utility-Projection (U) Encode each offer by its scalar utility. **✓** Generalizes. **✗** Discards *all* outcome-space structure.

Raw-Outcome (I/O) Encode offers one-hot / per-issue. **✓** Retains structure. **✗** Breaks when issues or utilities change.

The codec defines the *problem the policy solves*; it is not learnable.



WHERE EXISTING METHODS SIT

Method	$\mathcal{OE}/\mathcal{AD}$	Generalize	OS Struct.	No Feat. Eng.
Razeghi et al. (2020)	Ud / A	✓	✗	✓
RLBOA, Bakker et al. (2019)	Udr / Udr	✓	✗	✓
Sengupta et al. (2022)	Uc / Uc	✓	✗	✓
MiPN, Higa et al. (2023)	Id / Id	✗	✓	✓
VeNAS, Takahashi et al. (2022)	Od / Od	✗	✓	✓
Renting ⁺ et al. (2024)	Gd / Gd	✓	✓	✗
VUDO (Ours)	VUDO	✓	✓	✓

GENERALIZED PROCUREMENT

Buyer–seller, random GLA utilities. *Different* opponents in train (Atlas3, CUHK, Boulware, Conceder) and test (AgentK, Hardheaded, AgentGG, Linear). Raw-Outcome methods are inapplicable here.

Method	Advantage	Welfare	Fairness (Nash)
Atlas3	0.154 (0.279)	0.297 (0.375)	0.425 (0.284)
Hardheaded	0.176 (0.354)	0.240 (0.307)	0.340 (0.179)
AgentK	0.201 (0.380)	0.259 (0.299)	0.349 (0.178)
AgentGG	0.174 (0.348)	0.238 (0.298)	0.345 (0.174)
Sengupta	0.191 (0.328)	0.323 (0.379)	0.448 (0.292)
RLBOA ⁺	0.160 (0.306)	0.279 (0.349)	0.389 (0.251)
Renting ⁺	0.230 (0.253)	0.378 (0.277)	0.475 (0.214)
VUDO	0.298 (0.272)	0.683 (0.329)	0.678 (0.239)
VUDONoAct	0.215 (0.319)	0.376 (0.407)	0.497 (0.325)
VUDONoObs	0.223 (0.342)	0.417 (0.393)	0.467 (0.254)

VUDO wins **advantage, welfare, fairness** (Wilcoxon, Bonferroni-corrected). Ablations confirm the codec helps on *both* sides.

DESIGN PRINCIPLE

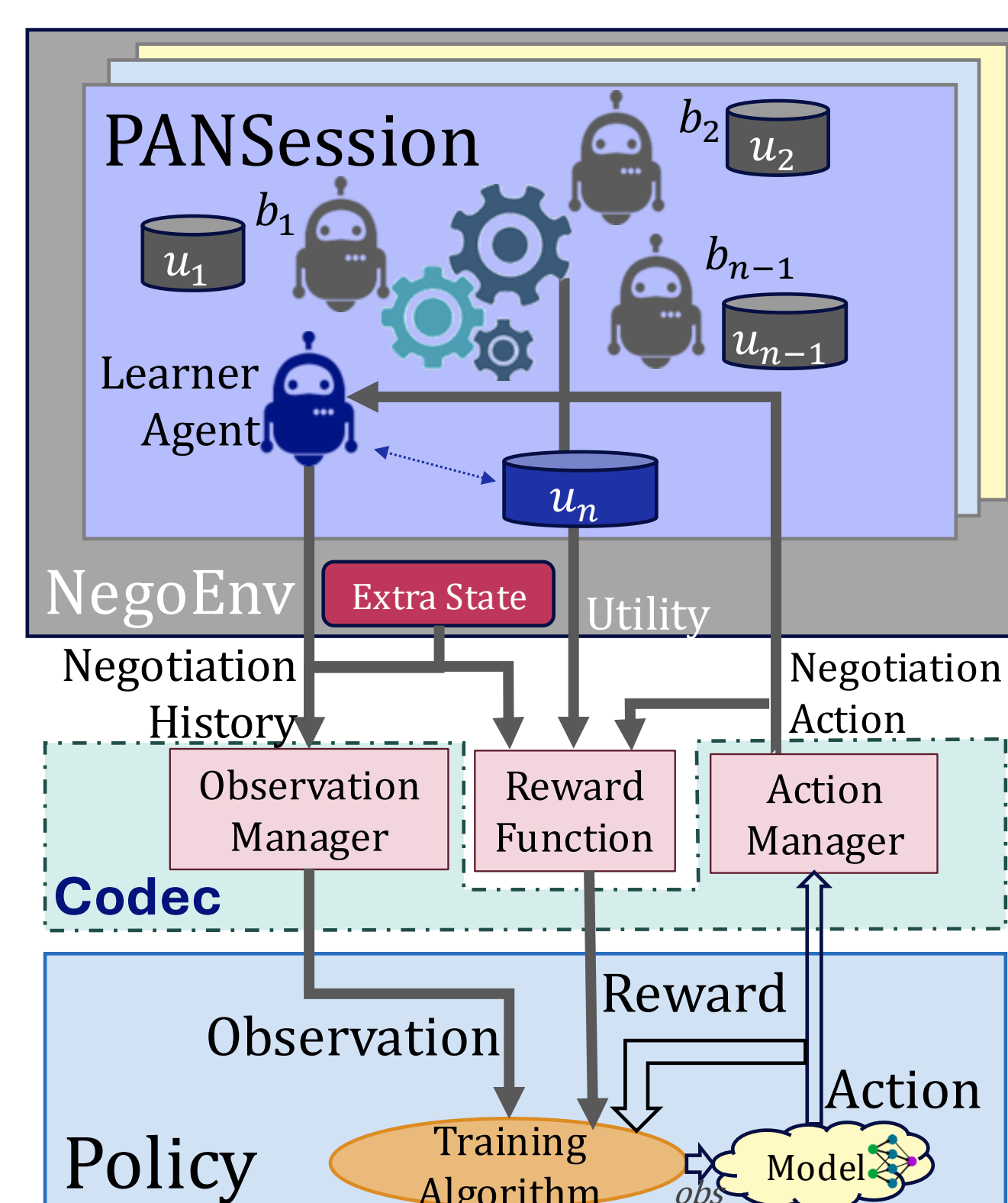
Two outcomes with the same value-function vector for *all* agents are **negotiation-equivalent**: they share utilities, ranks, and relations to every other outcome – only this combination matters for the policy.

So encode an outcome by its **(weighted) value-function vector** – not by raw identity, not by scalar utility. This keeps everything the policy needs to act, while remaining **invariant** to outcome-space size and to how issues are labeled or grouped.

FRAMEWORK: NEGMAS-RL

An RL for AN problem is a tuple $(\mathcal{D}^{trn}, \mathcal{C} \equiv (\mathcal{OE}, \mathcal{AD}), \mathcal{R}, \mathcal{F})$ that induces a finite-horizon POMDP.

- **PGP** factors $P(\mathcal{P}) P(\lambda | \mathcal{P}) P(\mathcal{B} | \cdot) P(\mathcal{I} | \cdot)$ – sample protocols, scenarios, backgrounds and assignments independently.
- **$\mathcal{OE}$** defines the observation space $\mathcal{O}_m$ from negotiation history.
- **$\mathcal{AD}$** defines the action space $\mathcal{A}_m$ (offer / accept / withdraw).
- **$\mathcal{R}, \mathcal{F}$** defaults to advantage at agreement; shaping is optional.
- Swap $(\mathcal{OE}, \mathcal{AD}, \mathcal{R}, \mathcal{F})$ to recover SOTA methods – the training loop never changes.



Expresses RLBOA, Sengupta, VeNAS, MiPN, Renting⁺, ... as different $(\mathcal{OE}, \mathcal{AD}, \mathcal{R}, \mathcal{F})$ on the same PGP.

PROPOSED CODEC: VUDO

Encoder. Project ω to $x = \langle \alpha_k \nu_k(\omega_{g_k}) \rangle_{k=1}^K$, then split heaviest / merge lightest weights to reach a fixed target dimensionality d . Same projection for own offers and the opponent's, so the policy sees a uniform d -vector regardless of the underlying outcome space.

Decoder. Project the outcome space once per scenario and cache it; map a policy output back by nearest-neighbour (cosine) – approximate NN search scales to large Ω .

- ✓ Generalizes
- ✓ No feature engineering
- ✓ Architecture-agnostic

ANAC 2024 SCENARIOS

100 training / 50 test scenarios from the ANAC 2024 competition.

Method	Advantage	Welfare	Fairness (Nash)
Atlas3	0.163 (0.300)	0.222 (0.398)	0.522 (0.209)
AgentK	0.153 (0.350)	0.158 (0.362)	0.454 (0.140)
AgentGG	0.040 (0.196)	0.040 (0.196)	0.448 (0.133)
Hardheaded	0.040 (0.196)	0.040 (0.196)	0.448 (0.133)
RLBOA ⁺	0.077 (0.141)	0.212 (0.367)	0.504 (0.186)
Sengupta	0.117 (0.251)	0.887 (0.123)	0.529 (0.149)
Renting ⁺	0.018 (0.120)	0.451 (0.246)	0.635 (0.204)
VUDO	0.179 (0.323)	0.227 (0.406)	0.511 (0.196)
VUDONoAct	0.048 (0.111)	0.334 (0.429)	0.495 (0.155)
VUDONoObs	0.050 (0.016)	0.891 (0.123)	0.481 (0.113)

VUDO leads all RL baselines on advantage. On the easy single-scenario / Boulware setup (table omitted for space) only AgentK is statistically indistinguishable. VUDO costs only **+3.5%** parameters and **+1.5%** training time over Sengupta; Renting⁺ is **~5.3×** slower.

TAKEAWAYS

- **Codec** = a lever that hugely influences RL-negotiator generalization.
- VUDO keeps **all** negotiation-relevant info while staying **scenario-agnostic**.